A set is computably enumerable iff it is the projection of a computable relation

The Open Logic Project contributors

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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel thm:exists-char of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- enumerable sets as W_e, the domain-range description of enumerations
- Kleene's T predicate
- unbounded minimization producing partial computable functions

Theorem 0.1. *A set S is computably enumerable if and only if there is a computable relation $R(x, y)$ such that*

$$S = \{ x \mid \exists[y][R(x, y)] \}.$$

Proof. In the forward direction, suppose S is computably enumerable. Then for some e, $S = W_e$. For this value of e we can write S as

$$S = \{ x \mid \exists[y][T(e, x, y)] \}.$$

In the reverse direction, suppose $S = \{ x \mid \exists[y][R(x, y)] \}$. Define f by

$$f(x) \simeq \mu y R(x, y).$$

Then f is partial computable, and S is the domain of f. □

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